

P8109 Statistical Inference – Homework 3

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Problem 1

Let X_1 and X_2 be two independent Poisson random variables with parameter λ . Let $T = X_1 + 2X_2$.

(a) By finding an expression for $P(X_1 = 1, X_2 = 1 \mid T = 3)$, show that T is not sufficient for λ .

Since $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$, by independence,

$$P(X_1 = x_1, X_2 = x_2) = \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} = \frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!}.$$

Therefore,

$$P(X_1 = 1, X_2 = 1 \mid T = 3) = \frac{P(X_1 = 1, X_2 = 1)}{P(X_1 = 1, X_2 = 1) + P(X_1 = 3, X_2 = 0)} = \frac{e^{-2\lambda} \lambda^2}{e^{-2\lambda} \lambda^2 \left(\frac{\lambda}{6} + 1\right)} = \frac{1}{\frac{\lambda}{6} + 1} = \frac{6}{\lambda + 6}.$$

Since the expression depends on λ , we conclude that $T = X_1 + 2X_2$ is **not** sufficient for λ .

(b) By using the factorization theorem, obtain a sufficient statistic for λ .

The joint likelihood of (X_1, X_2) is

$$L(\lambda \mid x_1, x_2) = \prod_{i=1}^2 \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!} = \underbrace{e^{-2\lambda} \lambda^{x_1+x_2}}_{g_\lambda(S)} \cdot \underbrace{\frac{1}{x_1! x_2!}}_{h(x_1, x_2)},$$

By the Fisher–Neyman Factorization Theorem (Theorem 5.1),

$$S = X_1 + X_2$$

is a sufficient statistic for λ .

Problem 2

Consider a discrete random variable X with pmf $p(x; \theta)$, $x = 1, 2, 3$, $\theta = \theta_1, \theta_2, \theta_3$, given by:

x	θ_1	θ_2	θ_3
1	0.1	0.2	0.3
2	0.7	0.4	0.1
3	0.2	0.4	0.6

Let $T = 0$ if x is odd, $T = 1$ if x is even.

(a) Obtain $p(x | T; \theta)$ in tabular form for $T = 0$ and $T = 1$.

Using $p(x | T; \theta) = p(x; \theta) / P(T; \theta)$, with $P(T = 0; \theta) = p(1; \theta) + p(3; \theta)$:

$T = 0$ ($x \in \{1, 3\}$):

x	θ_1	θ_2	θ_3
1	$0.1/0.3 = 1/3$	$0.2/0.6 = 1/3$	$0.3/0.9 = 1/3$
3	$0.2/0.3 = 2/3$	$0.4/0.6 = 2/3$	$0.6/0.9 = 2/3$

$T = 1$ ($x = 2$ is the only even value, so trivially):

x	θ_1	θ_2	θ_3
2	1	1	1

(b) Comment on the nature of T .

Both conditional tables in (a) are free of θ . By Definition 5.1, T is **sufficient** for θ .

Problem 3

Let $X_1, \dots, X_n$ be i.i.d. with pdf $f(x; \theta) = \theta^{-1}$ for $|\theta| < x < 2\theta$, and 0 elsewhere. Obtain a sufficient statistic for θ .

The support $(|\theta|, 2\theta)$ is non-empty only if $\theta > 0$, so $|\theta| = \theta$ and the support is $(\theta, 2\theta)$. The joint likelihood is

$$L(\theta | \mathbf{x}) = \theta^{-n} \prod_{i=1}^n \mathbf{1}(\theta < x_i < 2\theta) = \theta^{-n} \underbrace{\mathbf{1}\left(\frac{x_{(n)}}{2} < \theta < x_{(1)}\right)}_{g_\theta(x_{(1)}, x_{(n)})} \cdot \underbrace{1}_{h(\mathbf{x})},$$

since $\theta < x_i$ for all i iff $\theta < x_{(1)}$, and $x_i < 2\theta$ for all i iff $x_{(n)} < 2\theta$. By the Factorization Theorem, $\mathbf{T} = (X_{(1)}, X_{(n)})$ is sufficient for θ .

Problem 4

Let $X_1, X_2 \stackrel{\text{iid}}{\sim} N(\theta, 1)$. Let $d = (X_1 + X_2)/2$ and $d^* = E(d | X_1)$.

(a) Obtain the expectations of d and d^* .

By linearity, $E(d) = (\theta + \theta)/2 = \theta$.

Since $X_1 \perp X_2$, we have $E(X_2 | X_1) = \theta$, so

$$d^* = E(d | X_1) = \frac{X_1 + E(X_2 | X_1)}{2} = \frac{X_1 + \theta}{2}.$$

By the tower property, $E(d^*) = E[E(d | X_1)] = E(d) = \theta$.

(b) Show that $\text{Var}(d^*) < \text{Var}(d)$.

$$\text{Var}(d) = \frac{\text{Var}(X_1) + \text{Var}(X_2)}{4} = \frac{1}{2}, \quad \text{Var}(d^*) = \frac{\text{Var}(X_1)}{4} = \frac{1}{4} < \frac{1}{2} = \text{Var}(d).$$

(c) Explain why d^* cannot be used as an improved estimator of θ .

$d^* = (X_1 + \theta)/2$ depends on the unknown θ , so it is **not a valid statistic** and cannot be computed from data. Moreover, X_1 is not sufficient for θ (the sufficient statistic is $X_1 + X_2$ by the factorization theorem), so this does not constitute a Rao–Blackwell improvement (Theorem 5.3).

Problem 5

Show that the $\text{Uniform}(\theta, \theta + 1)$ family is not complete.

Let $h(x) = \sin(2\pi x)$, which is not identically zero. For $X \sim \text{Uniform}(\theta, \theta + 1)$:

$$E_\theta[h(X)] = \int_\theta^{\theta+1} \sin(2\pi x) dx = \left[-\frac{\cos(2\pi x)}{2\pi} \right]_\theta^{\theta+1} = \frac{\cos(2\pi\theta) - \cos(2\pi(\theta + 1))}{2\pi} = 0,$$

since $\cos(2\pi(\theta + 1)) = \cos(2\pi\theta)$. We have exhibited a non-zero h with $E_\theta[h(X)] = 0$ for all θ , so by Definition 6.2 the family is **not complete**.

Problem 6

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Binomial}(m, p)$. We wish to obtain the UMVUE of $p^r q^s$, where $q = 1 - p$ and r, s are known positive integers with $r + s < mn$.

(a) Obtain the distribution of $T = X_1 + \dots + X_n$.

The MGF of $X_i \sim \text{Binomial}(m, p)$ is $M_{X_i}(t) = (q + pe^t)^m$. By independence,

$$M_T(t) = \prod_{i=1}^n M_{X_i}(t) = (q + pe^t)^{mn},$$

which is the MGF of $\text{Binomial}(mn, p)$. By uniqueness, $T \sim \text{Binomial}(mn, p)$.

(b) Show that $U(T) = \binom{mn-r-s}{T-r} / \binom{mn}{T}$ for $T = r, \dots, mn-s$ is unbiased for $p^r q^s$.

Expand the right-hand side using $(p+q)^{mn-r-s} = 1$:

$$p^r q^s = \sum_{j=0}^{mn-r-s} \binom{mn-r-s}{j} p^{j+r} q^{mn-(j+r)}.$$

Substituting $t = j + r$:

$$p^r q^s = \sum_{t=r}^{mn-s} \binom{mn-r-s}{t-r} p^t q^{mn-t}.$$

Since $T \sim \text{Binomial}(mn, p)$, we need $\sum_t U(t) \binom{mn}{t} p^t q^{mn-t} = p^r q^s$. Matching coefficients of $p^t q^{mn-t}$ gives

$$U(t) = \frac{\binom{mn-r-s}{t-r}}{\binom{mn}{t}}, \quad t = r, \dots, mn-s, \quad U(t) = 0 \text{ otherwise.}$$

(c) Argue briefly that $U(T)$ is the UMVUE of $p^r q^s$.

The Binomial(m, p) family is a one-parameter exponential family of full rank, so by the Factorization Theorem $T = \sum X_i$ is sufficient, and by Theorem 6.2 T is complete. Since $U(T)$ is unbiased for $p^r q^s$ (part (b)) and is a function of the complete sufficient statistic T , the Lehmann–Scheffé Theorem (Theorem 6.3) implies $U(T)$ is the UMVUE of $p^r q^s$.